

Biology

Chapter 1: Characteristics and Classification of Living Organisms

Welcome to AnithUncommon's **Sample Resources!**

In Biology O'level, 1.1...

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Yassmin Younes to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology O'Level, join our nonprofit!

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This sample is made thanks to, [Yassmin Younes](#).

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- List and describe the seven characteristics of living organisms and explain their importance in defining "life."
- Describe the features of the five kingdoms.
- Describe the features of viruses and explain why they are generally not considered living organisms.
- Identify and classify known organisms using dichotomous keys based on characteristic features.

Reflective Questions to Consider:

Organisms are the diverse building blocks of our biosphere. While a giant redwood tree and a microscopic bacterium appear to have nothing in common, they both perform the same fundamental processes to stay alive.

Below are the following core questions you will face as we explore this chapter:

- If a virus possesses genetic material and can evolve but cannot reproduce without a host, does it truly cross the threshold of "living," or is it merely a complex piece of biological machinery?
- How do the distinct structural differences between a prokaryote and an animal cell reflect an evolutionary "trade-off" between simplicity and complexity?
- How would science fall apart if we didn't have a "universal ID" for every living thing on Earth?

Syllabus Outline:

Unit 1: Characteristics and Classification of Living Organisms

Tip!

- You might struggle to remember all seven characteristics or feel confused when DNA evidence contradicts how an animal actually looks. To master this, focus on the "why" behind the groupings rather than just memorizing lists, and use the binomial system as a universal "ID card" to keep the vast diversity of species organized.

Activities in Class:

- Lesson Breakdown
- Hand-craft and label a detailed diagram of a virus
- Create a set of visual flashcards to master the diagnostic features of myriapods, insects, arachnids, and crustaceans
- Work in pairs to design a custom dichotomous key using mystery organisms.

1.1 Characteristics of organism

The Seven Characteristics (MRS GREN)

a. Movement → An action by an organism or part of an organism causing a change of position or place. → While animals move their entire bodies to find food or escape danger, plants move more subtly—such as turning their leaves toward the light (phototropism).

b. Respiration → The chemical reactions in cells that break down nutrient molecules and release energy for metabolism.

→ **Note:** Do not confuse this with "breathing." Respiration is a cellular process; think of it as the "engine" of the cell burning fuel to create power (ATP).

c. Sensitivity → The ability to detect or sense stimuli in the internal or external environment and to make appropriate responses. → Examples include your pupils constricting in bright light or a Venus flytrap closing when an insect touches its sensory hairs.

d. Growth → A permanent increase in size and dry mass by an increase in cell number or cell size or both. → This isn't just "getting bigger" temporarily (like a sponge soaking up water); it is a structural, lasting change involving the creation of new biological material.

e. Reproduction → The processes that make more of the **same kind** of organism. → This ensures the survival of the species over generations, whether through **asexual** (one parent) or **sexual** (two parents) means.

f. Excretion → Removal from organisms of the waste products of metabolism (chemical reactions in cells including respiration), toxic materials, and substances in excess of requirements.

Mentor's Notes:

In the sections below, you will see how these traits are interconnected. For example, Nutrition provides the raw materials for Respiration; Respiration then releases the energy (ATP) required for Movement and Growth. Meanwhile, excretion ensures that the toxic by-products of these chemical reactions don't build up and kill the organism.

Even if an organism is "rooted" like a plant, it moves by orienting its leaves toward the sun.

Note: Respiration is the most misunderstood term. It is NOT breathing; it is the chemical process of unlocking energy from food.

→ **Key Distinction:** Excretion is removing waste made *inside* your cells (like or urea). **This is different from egestion (passing out undigested food as feces).**

g. Nutrition → Taking in of materials for energy, growth, and development. → Plants require light, magnesium, and water to make their own food (**autotrophic**), while animals need organic compounds and ions which they usually eat (**heterotrophic**).

→ **Sensitivity is how life "navigates"—whether it's a bacteria moving away from poison or a human blinking at a bright light.**

Is there different ways of Nutrition?

→ **Unlike animals, Plants are the ultimate "chefs"—they make their own food using light. Animals are "consumers"—they must find and eat their nutrients.**

1.2 The Biological Classification System

The Binomial System

a. Definition

→ An internationally agreed system in which the scientific name of an organism is made up of two parts.

→ Think of the Genus as a "Surname" (shared by family) and the Species as a "First Name" (unique to the individual).

b. Structure

→ **Genus:** The first part of the name, always **capitalized** (e.g., *Panthera*).

→ **Species:** The second part of the name, always **lowercase** (e.g., *leo*).

→ **Benefit:** It prevents confusion caused by local "common names." A "mountain lion," "puma," and "cougar" are all the same animal: *Puma concolor*.

Defining a Species

a. The Fertility Rule

→ A Species is a group of organisms that can reproduce to produce fertile offspring.

→ If two different species mate (like a horse and a donkey), they might produce a mule, but that mule will be **infertile (cannot have babies)**. Therefore, the parents are different species.

Note how these concepts—Binomial Nomenclature, Genus, and Species—act as a universal language.

Try to RECOGNIZE that DNA sequencing is the "gold standard" in modern exams. If a question asks why a classification has changed, the answer is almost always that DNA analysis revealed a more accurate evolutionary relationship than just looking at physical features!

→ Think of the Genus as a "Surname" (shared by family) and the Species as a "First Name"

→ Binominal system prevents confusion caused by local "common names." A "mountain lion," "puma," and "cougar" are all the same animal: *Puma concolor*. Using the binominal naming system makes it easier for scientist to address these animals.

Methods of Classification

a. Morphology and Anatomy

→ **Morphology:** The study of the outward **shape** and **external** features of an organism.

→ **Anatomy:** The study of **internal** structures (like bone arrangements or organ systems).

→ **The Flaw:** Sometimes unrelated animals look similar because they live in the same environment (e.g., sharks and dolphins).

b. DNA and Protein Sequences

→ This is the most accurate way to classify organisms today.

→ Scientists compare the **sequences of bases in DNA** and the **sequences of amino acids in proteins**.

→ **The Rule:** The more similar the DNA sequences, the more closely related the organisms are, and the more recently they shared a common ancestor.

**Keep in mind that
Sometimes unrelated
animals look similar
because they live in
the same environment.**

1.2 Dichotomous Keys

The Concept of a Key

Definition → A "Dichotomous" key is a scientific tool used to **identify organisms** based on their physical traits.

→ It consists of a series of paired, contrasting statements (couplets) that lead the user to the correct name.

b. Function → To provide a standardized way for anyone to identify an unfamiliar species.

→ It relies on **morphology** (outward shape) and **anatomy** (internal structure) that can be easily seen.

Building and Using a Key

The Two-Way Choice → Every step offers exactly two options

→ One choice will lead you to a name, while the other will tell you which numbered step to go to next.

b. Choosing "Reliable" Features → **Good Features:** Number of legs, presence of antennae, leaf shape, number of petals, or wing venation.

→ **Bad Features:** Size (it's too small), Age (it's young), or Behavior (it likes to hide). These are not permanent or easily seen.

Types of Key Formats

a. The Branching Diagram (Spider Key) → Looks like a family tree where each branch splits into two. → Best for visualizing the "path" of identification.

Note how these keys rely on constant, observable features rather than vague descriptions.

We use these keys because they turn the chaos of nature into a structured, step-by-step process that anyone can follow to reach the same scientific conclusion.

Try to RECOGNIZE that Dichotomous Keys in exams usually focus on "observable features."

Never use "it is small" or "it is fast" as a distinguishing feature in your own keys, because these are relative. Instead, use "has 6 legs" or "has spots on its wings"—things that don't change regardless of who is looking at the specimen!

1.4 The Five Kingdoms of Life

Kingdom Classification

The Sorting Criteria:

→ **Cell Type:** Prokaryotic (no nucleus) vs. Eukaryotic (has a nucleus).

→ **Organization:** Unicellular (one cell) vs. Multicellular (many cells).

→ **Nutrition:** Autotrophic (makes own food) vs. Heterotrophic (eats others) vs. Saprophytic (absorbs dead matter).

The Five Kingdoms Breakdown

a. Animals (Animalia) → Multicellular eukaryotes that lack cell walls and chloroplasts. → They are heterotrophs that usually move their whole bodies to find food.

b. Plants (Plantae) → Multicellular eukaryotes with cellulose cell walls and chloroplasts. → They are autotrophs that use chlorophyll to trap light energy for photosynthesis.

c. Fungi → Multicellular (except yeast) eukaryotes with cell walls made of **chitin**. → They lack chloroplasts and feed by saprophytic nutrition (absorbing nutrients from decaying matter).

d. Protoctists (Protoctista) → eukaryotes that are mostly unicellular (like Amoeba). → Some have chloroplasts and cell walls (like algae), while others behave more like animals.

e. Prokaryotes (Prokaryotae / Bacteria) → Microscopic, unicellular organisms that **lack a nucleus** and membrane-bound organelles. → They have cell walls made of **peptidoglycan** and contain circular DNA (plasmids).

Note how these kingdoms are defined by three main questions: Is it made of one cell or many? Does it have a nucleus? And how does it get its food?

We use this system because it allows us to organize millions of species into manageable, logical clusters based on their fundamental design.

Try to RECOGNIZE the difference between Fungi and Plants.

While they both stay in one place and have cell walls, Fungi never have chloroplasts and never photosynthesize.

When studying the Prokaryote, Kingdom,

remember your cell notes:

Prokaryotes are the only Kingdom that lacks a nucleus.

If you see a cell diagram with loose, circular DNA and no nuclear envelope, you can instantly classify it as a Prokaryote!

Feature	Animal	Plant	Fungus	Protocist	Prokaryote
Nucleus	✓	✓	✓	✓	✗
Cell Wall	✗	✓ (Cellulose)	✓ (Chitin)	Varies	✓ (Peptidoglycan)
Chloroplasts	✗	✓	✗	Varies	✗
Organization	Multicellular	Multicellular	Mostly Multi	Mostly Uni	Unicellular
Nutrition	Ingestive	Photosynthetic	Saprophytic		

1.5 Groups within the Plant and Animal Kingdoms

I. The Animal Kingdom: Vertebrates

Animals with a backbone (the Phylum Chordata). There are five main classes:

a. Mammals → Have hair or fur; have a placenta; young are fed on milk from mammary glands; endothermic (warm-blooded).

b. Birds → Have feathers; have a beak; front limbs are modified into wings; lay hard-shelled eggs; endothermic.

c. Reptiles → Have dry, scaly skin; lay rubbery-shelled eggs on land; ectothermic (cold-blooded).

d. Amphibians → Have moist, scale-less skin; lay jelly-like eggs in water; larvae (tadpoles) have gills, adults have lungs.

e. Fish → Have scales; have fins; breathe through gills; lay jelly-like eggs in water.

Just as different organelles give a cell its unique function, these structural features define the "vocation" of an organism.

Try to RECOGNIZE the "Rule of Pairs" when identifying Arthropods. Insects always have 3 pairs of legs, while Arachnids have 4 pairs. If you see 5 or more pairs, you are looking at a Crustacean or a Myriapod.

When differentiating between Monocotyledons

II. The Animal Kingdom: Invertebrates (Arthropods)

Animals without a backbone, but with jointed legs and an exoskeleton.

Group	Legs	Body Segments	Other Features
Insects	3 pairs	3 (Head, thorax, abdomen)	1 pair of antennae; usually 2 pairs of wings
Arachnids	4 pairs	2 (Cephalothorax, abdomen)	No antennae; simple eyes
Crustaceans	5+ pairs	2 (Cephalothorax, abdomen)	2 pairs of antennae; often have claws
Myriapods	Many pairs	Many segments	1 pair of antennae; includes centipedes/millipedes

III. The Plant Kingdom

Plants are classified by how they reproduce and their leaf structure.

a. Ferns → Do not produce seeds; they reproduce via **spores** found on the underside of their leaves (fronds).

and Dicotyledons, look at the leaf veins. Parallel veins mean "Monocot," while branching (net-like) veins mean "Dicot." It is a visual shortcut that never fails in identification questions.

Why these specific groups?

Because they represent the major evolutionary "upgrades" in history. The move from being a simple invertebrate to a vertebrate with a backbone, or the development of flowering plants from simple ferns, changed how life occupied the planet.

b. Flowering Plants → Reproduce using seeds formed inside flowers. They are split into two groups:

Feature	Monocotyledons (Monocots)	Dicotyledons (Dicots)
Seeds	One cotyledon (seed leaf)	Two cotyledons
Leaf Veins	Parallel veins	Branching (net-like) veins
Flowers	Parts in multiples of 3	Parts in multiples of 4 or 5
Roots	Fibrous root system	Taproot system

1.6 Viruses: The Non-Cellular Particles

Basic Structure

Unlike the eukaryotic and prokaryotic cells we have studied, a virus is incredibly simple. It consists of only two primary components:

a. Genetic Material → A core of either **DNA or RNA** (never both).
→ This contains the instructions needed to make more viruses once inside a host.

b. Capsid (Protein Coat) → A protective outer layer made of protein that surrounds the genetic material.

→ Some viruses also have an extra outer "envelope" stolen from the host's cell membrane, but the capsid is the universal feature.

Try to **RECOGNIZE** that a virus is much, much smaller than a bacterium.

Be careful with the term **"Reproduction."**
Viruses do not reproduce; they are replicated by the host cell.

Mentors' Sources:

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